

Pratap Singh Thakur

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Professional Summary

Computer Science graduate with a strong foundation in programming, problem-solving, and analytical thinking. Experienced in building full-stack web applications using the MERN stack and working with databases, REST APIs, and AI-powered systems. Skilled in building Retrieval-Augmented Generation (RAG) applications using LLMs, vector databases, and embeddings. Passionate about Artificial Intelligence, Large Language Models (LLMs), and evaluating AI systems. Eager to contribute to AI research while continuously learning emerging technologies.

Skills

- Programming: Java, JavaScript, C, C++, SQL
- Backend: Node.js, Express.js, REST APIs
- Frontend: React.js, HTML5, CSS3, Tailwind CSS
- Version Control: Git, GitHub
- Databases: MongoDB, MySQL
- Deployment: Vercel, Render
- Tools: Postman, VS Code
- AI/ML: LangChain, OpenAI/Claude API, RAG, Vector Embeddings, Prompt Engineering

Projects

InterviewAI — Agentic AI Interview Platform GitHub

- Built an AI interview platform using Next.js, LangGraph, ChromaDB, and RAG for automated technical and HR interview evaluation workflows.
- Integrated RAG with ChromaDB to parse candidates' resumes against job descriptions, enabling dynamic and contextual follow-up questions.
- Reduced average interview response latency by 35 percent using asynchronous multi-agent workflows with Redis caching and scalable LangGraph orchestration pipelines.
- Tech Stack: JavaScript, Node.js, Express.js, MongoDB, FastAPI, LangChain, OpenAI API, ChromaDB, React.js, Tailwind CSS, , , Vercel/Render

AI Resume Optimizer (ATS Analyzer) live

- Developed a full-stack web application using React.js, Node.js, Express.js, and MongoDB to analyze resumes against job descriptions.
- Built a responsive React.js frontend for resume upload, job description input, and visualization of ATS compatibility results.
- Designed and implemented REST APIs using Node.js and Express.js for resume processing, job description analysis, and result generation.
- Tech Stack: React.js, Node.js, Express.js, MongoDB, REST APIs, JavaScript, Text Similarity

E-Commerce Shopping Website Live

- Developed a full-stack e-commerce application using the MERN stack with user authentication, product management, cart, and order functionality.
- Implemented secure user authentication and authorization using JWT, protected routes, and Express.js middleware.
- Designed and developed RESTful APIs for user, product, cart, and order management using Node.js and Express.js.
- Built responsive and reusable React.js components for product browsing, cart management, and user interactions.
- Tech Stack: React.js, Node.js, Express.js, MongoDB, Mongoose, JWT, JavaScript, HTML, CSS, REST APIs

Education

VIT Bhopal University, B.Tech in Computer Science and Engineering CGPA: 7.39/10 Aug 2022 – May 2026

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks

Certifications

- AWS Certified Cloud Practitioner [Link] April 2026
- MERN FullStack – Ethnus [Link] Jan 2024

Achievements & Extracurricular Activities

- Participated in TCS Codevita Coding Contest And Cleared TCS CodeVita Season 13– Round 1 [Link]
- Solved 200+ Data Structures and Algorithms problems across LeetCode and GeeksforGeeks. [Link] [Link]
- Actively participated in technical competitions and coding challenges to sharpen problem-solving abilities.